RATING METHODOLOGY

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Rating Methodology

Midstream Energy

This rating methodology replaces the *Midstream Energy* methodology published on 25 February 2022. While this methodology reflects many of the same core principles as the 2022 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Scale factor, we have replaced the Property, Plant & Equipment, Net (PP&E) subfactor with Fixed Assets. In the Business Profile factor, we have renamed the Estimated Price and Volume Risk Exposure subfactor to Cash Flow Stability. In the Leverage and Coverage factor, we have removed the (FFO - Maintenance CAPEX) / Distributions subfactor. We have changed some factor weights and scoring thresholds. We have also made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies, and we have changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in midstream energy. In broad terms, midstream energy companies employ fixed infrastructure to connect energy production (upstream activities) to downstream markets, including petroleum refiners. The customers of midstream energy companies are mainly other energy companies.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

We use this methodology to rate companies that are primarily engaged in one or several aspects of the midstream energy business, including oil and natural gas gathering systems; pipelines that transport crude oil, petroleum products, natural gas or chemicals; natural gas processing, fractionation, liquefaction or re-gasification; or storage of energy and related products (including in-ground natural gas storage and tank farms).

This methodology is also applicable to holding companies with activities limited to owning minority interests in operating companies engaged in the midstream energy business. We typically expect companies that we rate using this methodology to have at least 15% direct or indirect ownership on a sustainable, forward-looking basis and to have some meaningful influence or control over decisions at the operating company. For details on specific credit considerations for these types of holding companies and guidance for scoring these companies using the midstream energy scorecard, please see the appendix. We typically rate holding companies that have minority equity stakes in more than three midstream operating companies using our methodology for investment holding companies and conglomerates.[1]

We rate certain subsectors within the midstream sector using other methodologies. We rate companies that generate at least two-thirds of their cash flow from natural gas long-haul transportation and ancillary activities that are tariff-regulated, or where contract terms are set within a tariff-regulated framework, using other methodologies. For details, see our methodologies for natural gas pipelines, regulated electric and gas utilities, and regulated electric and gas networks. We rate midstream energy project financings using the applicable project finance methodology, including our methodology for generic project finance.

We typically classify as natural gas pipelines those companies that generate at least two-thirds of their cash flow from natural gas pipelines and ancillary activities provided that they are not organized as a master limited partnership (MLP) or similar growth-oriented yield vehicle (yieldco) and do not own an MLP or yieldco. We rate natural gas pipelines using our natural gas pipelines methodology.

Although certain midstream energy companies have some exploration and production, petroleum refining or marketing as part of their business mix, we rate companies that are primarily engaged in these activities using other methodologies. For details, see our methodologies for independent exploration and production companies, integrated oil and gas companies, and refining and marketing companies. We also rate companies that connect upstream and downstream energy markets via surface or water-borne transportation using other methodologies. For details, see our methodologies for shipping and for surface transportation and logistics.

Sector overview

The midstream sector comprises companies with varying levels of scale, asset types and business mixes. Different hydrocarbon products (e.g., natural gas, natural gas liquids, crude oil or petroleum products) require different types of infrastructure. Also, the business profiles of midstream energy companies vary greatly, ranging from higher-risk activities with volume and commodity price exposure (e.g., natural gas processing, marketing and trading) to lower-risk, long-term contracted assets and fee-based services (e.g., crude oil pipelines, gas pipelines and storage).

Leverage has been a defining characteristic for many companies in the sector, partly because of the stable business profiles of many companies and partly because of the eligibility of these assets in certain jurisdictions for tax-advantaged partnership structures that require high levels of cash flow distribution. Certain other companies, while not structured as partnerships, have adopted the business model of high cash payouts as a means to attract equity capital. The sector has historically had an orientation toward growth that is often debt-financed, including organic expansion and mergers and acquisitions.

Rated midstream energy companies have predominantly been based in the US, and many are MLPs. The MLP structure incentivizes cash flow growth and high cash payouts, while offering tax advantages and access to public equity markets. It consists of limited partners (LPs), who hold publicly traded units, and a general partner (GP), who manages operations. The value of an MLP's equity is primarily driven by distribution growth, which places pressure on the partnership to sustain expansion through asset acquisitions or construction projects, often financed with debt.

MLPs do not pay U.S. federal income taxes if at least 90% of their earnings derive from qualifying natural resource activities. Instead, LPs are taxed on their share of distributed income, much of which is shielded by depreciation and amortization.

Under the terms of the partnership agreement, MLPs are required to distribute all "available" cash quarterly, typically all cash from operations after debt service, maintenance capital spending, and cash held in discretionary reserves. Because MLPs do not retain earnings and may find it difficult to grow organically, they require regular access to debt and equity markets to finance expansion. If growth slows, MLPs may pursue riskier business lines to maintain returns.

The MLP structure raises corporate governance concerns. Unitholders have limited liability but minimal voting rights, while control resides with the GP, who may extract value to the detriment of creditors. The GP may also hold incentive distribution rights, which entitle it to an increasing share of cash distributions as earnings targets are met — potentially up to 50%. As the GP's share rises, the MLP's cost of capital increases, intensifying the pressure to grow. Investment discipline is therefore a positive credit characteristic, as MLPs must regularly demonstrate the viability of growth projects to credit and equity markets.

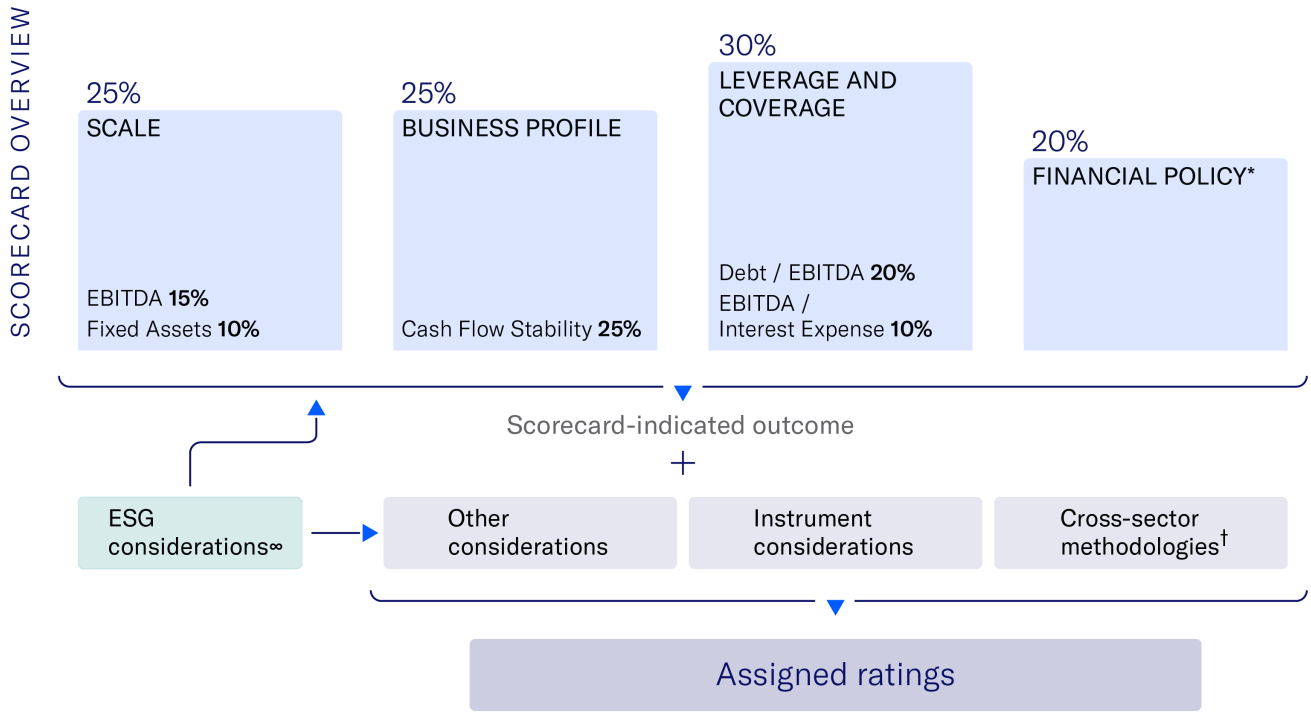
MLPs often carry high financial leverage and are not wholly owned by their sponsoring parent, which can lead to heightened structural subordination and complicate the assessment of consolidated financial metrics. The parent's ownership level affects the accounting treatment of the MLP's debt and adds complexity to evaluating the risk profile of the corporate family.

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the midstream energy rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Midstream energy scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (25%)									
EBITDA (USD Billion)[1]	15%	≥ $15	$10 - $15	$5 - $10	$1 - $5	$0.5 - $1	$0.1 - $0.5	$0 - $0.1	< $0
Fixed Assets (USD Billion)[2]	10%	≥ $80	$60 - $80	$40 - $60	$15 - $40	$5 - $15	$1 - $5	$0.5 - $1	< $0.5
Factor: Business Profile (25%)									
Cash Flow Stability	25%	Expected to have zero medium-to-long-term volume risk; no direct commodity price risk; strong commercial outlook or protected market; and high proportion of long-term contracts with highly rated counterparties.	Expected to have modest long-term volume risk but no medium-term risk; no direct commodity price risk; strong commercial outlook or protected market; and high proportion of medium- and long-term contracts with highly rated counterparties.	Expected to have limited medium-term volume risk; no direct commodity price risk; strong commercial outlook or protected market; high proportion of medium-term contracts with highly rated counterparties.	Expected to have modest near-to-medium-term volume risk; limited direct commodity price risk; strong commercial outlook.	Expected to have somewhat elevated volume risk; modest direct commodity price risk.	Expected to have substantial volume risk, substantial direct commodity price risk.	Expected to have very high price and volume risk.	Fully exposed to price and volume risk.
Factor: Leverage and Coverage (30%)									
Debt / EBITDA[3]	20%	≤ 0.5x	0.5x - 1.5x	1.5x - 3.5x	3.5x - 4.5x	4.5x - 5.5x	5.5x - 6.5x	6.5x - 7.5x	> 7.5x
EBITDA / Interest Expense[4]	10%	≥ 20x	15x - 20x	10x - 15x	5x - 10x	2.5x - 5x	1x - 2.5x	0.5x - 1x	< 0.5x

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Financial Policy (20%)									
Financial Policy	20%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions, or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $20 billion. A value of $20 billion or better equates to a numeric score of 0.5. The Ca endpoint value is $(0.05) billion. A value of $(0.05) billion or worse equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 10x. A value of 10x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[4] For the linear scoring scale, the Aaa endpoint value is 30x. A value of 30x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (25% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

Operations of larger midstream energy companies tend to weather temporary disruptions more effectively, owing to a generally broader mix of product and service offerings, wider geographical spread and greater exposure to producing basins. Larger companies may also derive economies of scale by allocating fixed costs across a broader asset base and securing more competitive supply procurement, ultimately supporting a stable or growing market position. These attributes are particularly meaningful in industries like midstream, which are heavily oriented toward cash-flow growth and acquisition-led expansion. Size may also enhance financial flexibility as larger companies typically benefit from easier access to capital markets and alternative financing sources.

How we assess it for the scorecard

We score this factor based on two quantitative subfactors: EBITDA and Fixed Assets.

EBITDA

We use earnings before interest, taxes, depreciation and amortization (EBITDA) in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Fixed Assets

We use gross property, plant and equipment (PP&E), expressed in billions of US dollars, to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (25% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

Different types of business operations within the midstream sector are subject to varying degrees of business risk. Operating assets characterized by low exposure to commodity price and volume risk, such as interstate refined products, natural gas or crude oil

pipelines, tend to generate predictable and stable long-term cash flows. Conversely, midstream energy companies that have expanded into upstream or downstream businesses may operate assets with relatively high exposure to commodity price and volume risk, making consistent cash flow stability more difficult to achieve.

Even within the same operating activity, the business risk of different companies (and in turn their cash flow volatility) can and often does vary significantly depending on the quality of operated assets, geographic location, or customer base. For example, a network of assets in a particular region, such as pipelines or contracted gathering systems, may hold dominant positions with producers or shippers, and competitors seeking to enter the region would have difficulty replicating the network economically. Certain assets may also have a critical position in a market, providing structural protection or effectively acting as an operational barrier to entry for potential competitors, all of which helps offset the intrinsic limitations of smaller scale. While larger scale generally confers flexibility or an operational competitive advantage, smaller midstream energy companies with a more limited range of operations may also demonstrate strengths in their business profile that may mitigate limited scale.

How we assess it for the scorecard

We score this factor based on one qualitative subfactor: Cash Flow Stability.

We score this factor based on a forward-looking assessment of the degree of stability in cash flows afforded by a company's contracts and market position as well as its estimated price and volume risk exposure.

Because midstream energy companies often operate various assets that are exposed to different price and volume risk, in our overall assessment we typically consider each of the individual business segments that may carry a singular risk profile. We then typically rank the underlying risk of all identifiable segments by the size of the segment to form an opinion of the company's overall business risk profile.

We typically consider, for a given company, the classification of business segments as reported by public sources and adjust it based on our view of the granular risks tied to the underlying assets in each of these segments. For example, we may break down a reported segment into subsegments (using reported or private information, or our own estimates) based on our view of material differences in the price and volume risk exposure of the various functions this segment encompasses.

We generally use EBITDA to estimate the relative importance of each business line. In cases where we consider that EBITDA does not accurately reflect the relative size and risks of a certain business line, we may use a different proxy, such as fixed assets.

For companies operating across different jurisdictions (across countries, states or provinces), we typically apply the same approach when gauging the regulatory and political risks a midstream energy company is exposed to. For example, in the US and Canada, intrastate and intra-provincial pipelines are subject to regulation at the state or provincial level, which can be more light-handed than national regulation. On the other hand, interstate pipelines in the US are regulated by the Federal Energy Regulatory Commission (FERC), which governs wholesale rates. The same applies to US intrastate pipelines with connections to FERC-regulated interstate pipelines. Ultimately, the varying regulatory frameworks carry different price and volume risks that may affect our overall assessment of business profile.

Our overall assessment of a company's business profile is not a point-in-time, sum-of-the-parts assessment, because even modest prospective exposure to higher-risk businesses, such as marketing and trading, can significantly raise a company's overall risk profile. As such, we ultimately make a qualitative assessment based on our view of a company's future strategic direction and underlying price and volume trends. Our assessment also focuses on the degree of perceived risk and on the time horizon for any materialization of this risk.

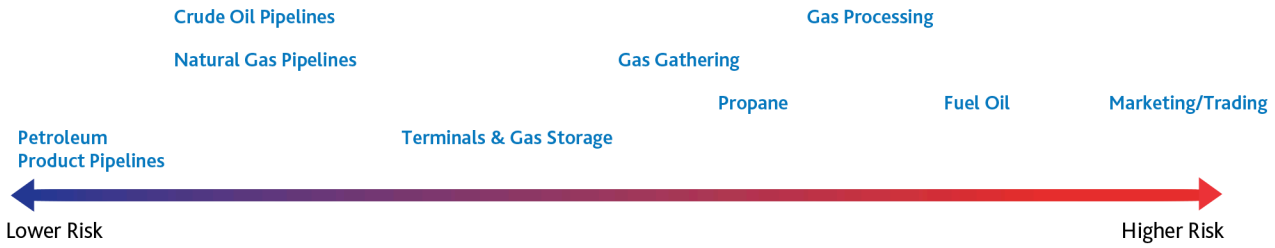
In the chart below and in the descriptions that follow it, we provide an indicative overview of price and volume risk exposures by business segment. The list encompasses the primary midstream business activities, but it is not exhaustive. The business profile score we assign ultimately depends on the specificities of each midstream activity that drive its price and volume risks. Specificities that bring differentiation in the degree of price and volume exposure include varying degrees of supply and demand fundamentals; the quality, tenor and structure of contracts; the strength and diversity of customers; the significance of market-driven revenues; geographic diversity; competition; and regulatory risk.

A midstream energy company characterized by a portfolio of assets that generates a contractually supported, fee-based cash flow stream, is regulated and faces minimal competitive challenges, is diversified by service type and geographic location, and operates in markets protected by de facto barriers to entry would typically be regarded as having a strong business profile. Fee-based contracts support a strong business profile because, under their terms, the processor typically takes limited or no commodity price risk and is compensated on a fee basis for volumes processed. Similarly, clauses limiting volume risk, such as minimum volume commitments (MVC), tend to support cash flow stability. For companies operating in sectors where volume or price movements are extensively driven by weather conditions (e.g., retail distribution), geographic diversity can help reduce sensitivity in any one market. Such a company is more likely to be able to support balance sheet leverage and relatively high cash distribution payouts.

Conversely, a midstream energy company whose asset base generates spot transactions unsupported by longer-term contracts can be expected to experience cash flow volatility that would not support balance sheet leverage and high distribution payouts. The company may be exposed to higher price and volume risk, if it operates in markets with minimal barriers to entry, faces intense competition, or lacks service or geographic diversification. In addition, market-based rate-setting or competitive pressure can lead to rate discounting or re-contracting risk.

When assessing exposure to commodity price risk, we generally consider the underlying price risk exposure, both direct and indirect. Indirect exposure relates to the impact commodity prices could have on customers or end markets, for example on drilling programs and hydrocarbon production. While hedging can mitigate direct exposure to commodity price risks, indirect risks typically remain.

Exhibit 3
The midstream risk spectrum



Note: For clarity, the illustration above includes many of the activities that midstream energy companies may have as part of their business mix. Companies that are primarily engaged in the distribution of propane and fuel oil and in marketing and trading, as well as companies that generate at least two-thirds of their cash flow from tariff-regulated natural gas pipelines, are rated using separate methodologies. Please see the "Scope" section for more details.
Source: Moody's Ratings

The following is an indicative overview of price and volume risk exposures by business segment:

Petroleum product pipelines: Petroleum product pipelines are typically considered the most stable of pipeline transportation assets, ranking as least risky along the business risk spectrum. Demand-driven petroleum product volumes tend to be highly stable, and they are not connected to depleting exploration and production (E&P) assets. High barriers to entry for new capacity, including rights-of-way, generally limit competition and promote volume stability. Product pipelines are generally subject to relatively light-handed regulation and less prone to regulatory risk than natural gas and crude oil pipelines. Contracts tend to be longer term, with contract rates typically adjusted annually on an inflation-indexed basis.

Natural gas pipelines: Natural gas pipelines are considered only slightly more risky than product pipelines because of their generic connection to natural gas reserves that can deplete over time. However, they typically remain relatively low risk, drawing the preponderance of their revenues from volume-insensitive demand charges typically set by regulation, under long-term contracts or a combination of the two.

Distinctions also exist between "demand pull" gas pipelines and "supply push" gas pipelines, particularly in terms of their customers. Demand pull pipelines transport natural gas primarily to commercial end users, such as gas distribution utilities and electric utilities with gas-fired generation. These customers typically exhibit a consistent track record of renewing their transportation contracts (and more often tend to have investment-grade ratings). Conversely, supply push pipelines, which transport natural gas out of producing

basins, tend to be of lower credit quality, with a wider dispersion of E&P contract counterparties, many of which tend to be speculative-grade-rated and are exposed to depleting resource bases.

In the US and Canada, intrastate and intra-provincial natural gas pipelines are considered incrementally riskier than federally regulated interstate pipelines. Compared with interstate or national pipelines, intrastate revenues generally are more sensitive to market conditions, particularly locational basis differentials, with rates negotiated with shippers on shorter term contracts.

Crude oil pipelines: Crude oil pipelines are typically considered relatively low-risk activities, with their positioning relative to natural gas pipelines generally dependent on market-specific outlooks for long-term production. In the US, rates and terms of service for natural gas and crude oil pipelines are regulated by FERC. Unlike gas pipelines, however, crude oil pipeline rates are set and adjusted according to an index set at the producer price index (PPI), allowing for rate predictability. Where growth projections for long-term crude oil production are softer than for natural gas, crude oil pipelines bear greater volumetric risk exposure, and vice versa. In North America, this has been the case in many regions for long periods. In all cases, the relative outlooks for oil and gas production can be region-specific and can change over time. For example, technological advances have enabled petroleum production from basins that were previously not economical to develop.

Terminals and natural gas storage: Natural gas storage in this section refers to unregulated gas storage, as opposed to regulated gas storage conducted by natural gas pipelines or distribution utilities. We generally consider terminal operations and natural gas storage as carrying moderate price risk, but proportionately greater volumetric risk. Contract terms are short to medium term, and utilization can be highly sensitive to local market conditions. Terminal and storage operations have often been extended further downstream to also include export facilities including crude oil and natural gas liquids (NGLs), and are typically integrated into a larger midstream asset base, providing them with greater cash flow stability and thereby diminishing the risk relative to standalone operations. We assess the terminalling and storage of non-hydrocarbon and other bulk commodities (which are typically unregulated) based on the commodity type, demand characteristics, geographic location, specific contract terms and volumetric exposure.

Natural gas gathering and processing (G&P): G&P can be a volatile business subject to commodity price and volume (throughput) risk. We consider gas processing to be the riskier component of G&P. Natural gas processing can be exposed to both volumetric risk and to direct commodity price risk. Different processing contract structures (keep-whole, percentage-of-proceeds and fee-based contracts) subject processing margins to varying degrees of commodity price risk, with keep-whole structures entailing the greatest exposure. In keep-whole contracts, the processor retains the NGLs removed from the gas stream and is obligated to reimburse the producer with gas equivalent in energy value, directly exposing processing margins to both NGL and gas price risk. Common in this sector, percentage-of-proceeds contracts also expose processors to commodity price risk in that all or a portion of their compensation is linked to the market price of the gas or liquids processed.

Natural gas gathering is the less risky component of G&P, because the business is predominantly fee-based. However, gas gathering is exposed to throughput risk and requires ongoing capital expenditures for well connections. While most gathering contracts are supported by either acreage dedications or minimum volume commitments (MVCs), acreage dedication offers little support if the producing field is not economical and not being drilled, or if it is in decline. Gathering systems are somewhat protected by competitive barriers to entry, because gathering systems, once in place and supported by acreage dedications, generally will not attract competing systems. By the same token, contractual MVCs tend to solve for volumetric risk but can render contracts not economical in an environment of low gas prices and can invite pressure to renegotiate or, in a producer bankruptcy, even contract rejection.

Regardless of contract structure, natural gas G&P is also exposed indirectly to commodity price risk, given the sensitivity of G&P volumes to natural gas prices. Producers are inclined to reduce their drilling investment when natural gas prices are weak. They are also inclined to minimize the percentage of their production that is processed when processing spreads are low, subject to ongoing compliance with pipeline content specifications. When ethane prices fall below natural gas on an energy-equivalent basis, liquids extraction becomes not economical. In this case, producers may reject ethane recovery, reducing overall NGL volumes.

We typically view G&P operations as having a more favorable impact on a company's business profile when they benefit from a strong competitive position in an established, low-cost field (or one with growing production) and when they are supported by a predominance of fee-based contracts. Having basin diversity, meaning discrete G&P assets operating across multiple producing basins,

is also helpful in mitigating price and volumetric risk. An absence of keep-whole processing contracts or contracts with terms that mitigate negative processing margins also contribute to a more favorable business profile.

By contrast, we typically view as less favorable small G&P operations concentrated in a single basin or a small set of fields, where there is a high degree of customer concentration, where production is declining or reinvestment needs are high, or where commodity price exposure is significant.

Propane and fuel oil distribution: Propane and fuel oil distributors are fragmented, highly seasonal businesses, with most of the demand for their product occurring during the winter heating season. Financial performance is typically a function of weather and thus relatively volatile and difficult to predict. Working capital needs can also be significant given the seasonal nature of consumption and the necessity of building inventories before the annual heating season. Of the two businesses, the fuel oil business tends to be more competitive and customer churn can be significant. Geographic diversity can help reduce sensitivity to weather or economic conditions in any one market.

Energy marketing and trading: Marketing and trading is at the high end of the risk spectrum because of the volatility of cash flow, the need for strong risk management and internal controls, margin risk and the potentially significant, sudden use of working capital for collateral posting. Volumes are difficult to predict, driven by commodity prices, weather and market technical factors. When gauging the relative risk of marketing and trading, we typically consider counterparty credit quality, the company's track record in trading execution, its liquidity resources, and its risk management capability, policies and procedures.

Factor: Leverage and Coverage (30% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on two quantitative subfactors: Debt/EBITDA and EBITDA/Interest Expense.[3]

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

EBITDA / Interest Expense

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to interest expense.

Factor: Financial Policy (20% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector offer attractive dividends to shareholders and operate with governance structures and financial strategies that significantly influence their credit profiles. In particular, MLPs often feature a separation between ownership and control, and this complexity can often lead to a company maintaining a less conservative financial policy. This is especially the case when the GP prioritizes yield-oriented strategies. Given the pressure to balance distributions with debtholders' interests, and the need for MLPs to access debt and equity markets, MLPs tend to have a higher risk profile than other midstream energy companies.

A midstream energy company's willingness to maintain or improve its credit profile can reflect a more disciplined approach to financial management. The ownership level of the GP and the simplicity of the partnership structure matter because substantial GP ownership

or the elimination of incentive distribution rights typically results in better alignment of interests with debtholders. Conversely, where publicly traded GPs carry their own debt obligations and rely primarily on distributions from the MLP for debt service, a leveraged GP may increase distribution pressure on the affiliated MLP or introduce event risk.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[4]

Carbon regulation and carbon transition represent, to varying degrees, a long-term risk for all companies involved in the production and processing of hydrocarbons. Certain aspects of the midstream energy sector have made it less exposed than some other oil and gas sectors, including long-term contracts that often insulate midstream energy companies from changes in commodity prices and volume risk. In addition, most midstream energy companies have considerable investments in infrastructure for natural gas, which has a lower carbon footprint than coal or oil. Nonetheless, carbon transition efforts will have a negative credit impact on the entire industry, which could increase counterparty credit risk for midstream energy companies. In addition, over the very long term, lower oil

and gas consumption will decrease the volumes that need to be transported and/or processed, which could exert negative pressure on midstream energy company credit quality. As for other long-term risks that may not be quantifiable, carbon transition risks are incorporated in our ratings analysis qualitatively, based on our view of trends that extend beyond the horizon for which more precise projections are practicable. As a result, in some circumstances, carbon transition risks may not be fully reflected in scorecard-indicated outcomes.

For companies in this sector, we also consider social issues that could materially affect the likelihood of default and severity of loss, for example through adverse impacts on business reputation and employee relations.

Audit committee financial expertise, the incentives created by executive compensation packages, related-party transactions, interactions with outside auditors, and ownership structure are among the areas we may consider in our assessment of how corporate governance affects a company's credit profile.

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[5]

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Rating general partners

We assess MLPs on a standalone basis as well as on a consolidated basis, including debt of the GP sponsor if it relies on distributions from the MLP to service its debt and to fund its own distributions. A corporate family rating (CFR) is typically assigned at the uppermost entity with rated debt in the legal organization. The debt instruments of speculative-grade GPs and their affiliates are rated in accordance with our loss given default cross-sector methodology.[6]

We typically rate a GP sponsor zero to three notches below the MLP, depending on the degree of explicit or implicit insulation between the GP and the MLP provided by the partnership agreement, financial policies or corporate governance mechanisms. We also typically consider whether the GP has its own assets that provide cash flow to service its debt.

For GPs without independent businesses that rely solely on distributions from the MLP, we typically rate the GP's debt two to three notches below the MLP's debt rating, acknowledging the leveraged nature of this cash flow stream. Although the GP's debt is typically non-recourse to the MLP, deterioration in the GP's credit quality could cause the MLP's ratings to be downgraded, although not necessarily in lock step.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity

stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

In this sector, regulatory considerations also play a role in our assessment of a company's cash flow stability. However, we may assess certain aspects of regulatory considerations outside of the scorecard, for instance when considering the effects on a company's credit quality if regulatory change is swift.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 4

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 5
Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 6
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Appendix: Holding companies owning solely a minority interest in a midstream operating company or companies

We may rate holding companies that own solely a minority interest in a midstream energy operating company, or a small number of such companies, using this methodology. We do so provided that the minority holding company (minority holdco) has some meaningful influence or control over decisions at the operating company (opco) or companies and typically at least a 15% direct or indirect ownership on a sustainable, forward-looking basis.

In these cases, a key component in our analysis is the rating of the opco or, in the absence of a rating, a standalone credit profile of the opco that a rating committee determines in accordance with this methodology. Additional analytical considerations include how we use the scorecard for the minority holdco, notching factors and other rating considerations.

Scoring factors for a minority holding company

For a minority holdco in the sector, we analyze its own credit characteristics and those of the opco using the factors and subfactors in the scorecard. We make adjustments as described below.

Factors: Scale and Business Profile

Because the minority holdco does not have operations distinct from its ownership interest in the opco, we typically use the opco's scores for the Scale and Business Profile factors and subfactors.

We assess Scale and Business Profile at the opco level. Where there are minority stakes in multiple opcos, we typically use composite scores for Scale and Business Profile based on the average of respective scores across opcos, weighted by the value of each ownership stake.

In estimating value for the weighted averages, we typically use the market value of the opco on a trading exchange, where available. If these are not available, we may use multiples of EBITDA or cash flow, transaction data from sales or acquisitions of comparable midstream energy companies, or other estimation techniques. Where we use a weighted average of a score, we round the resulting numeric weighted average to the nearest integer.

Factor: Leverage and Coverage

In assessing this factor for a minority holdco, we calculate or estimate proportionally consolidated metrics using minority holdco data and its proportional ownership interest of the opco.

We typically consider the cash flow generating ability of the opco, the minority holdco's proportionate share in the dividends or distributions paid by the opco, and the overall debt burden, including the additional leverage and debt service at the minority holdco level (including any intermediate holdcos). Another important consideration is the trajectory of dividends and distributions, including any potential that they could be restricted.

Debt / EBITDA

For debt, we use the sum of the minority holdco's proportionate share of the opco's debt and the debt at the minority holdco level, including any intermediate holdco debt. For EBITDA, we measure or estimate the distributions received by the minority holdco, which include any common and preferred dividends, limited and general partner distributions and interest on shareholder loans.

To assess this factor where there are minority stakes in multiple opcos, we typically score a composite metric calculated as the ratio of the sum of the numerators over the sum of the denominators. For example, if holdco A owns shares in opco B and opco C, we calculate Debt/EBITDA as:

1. holdco A's proportionate share of opco B debt plus holdco A's proportionate share of opco C debt plus holdco A's debt; divided by
2. holdco A's dividends received from opco B plus its dividends received from opco C

EBITDA / Interest Expense

For EBITDA, we measure or estimate the distributions received by the minority holdco, which include any common and preferred dividends, limited and general partner distributions and interest on shareholder loans. For interest expense, we measure or estimate the sum of the minority holdco's proportionate share of the opco's interest expense and the interest expense at the minority holdco level.

Factor: Financial Policy
We assess this factor for the minority holdco based on our view of risks related to financial policies at both the minority holdco and opco. We cap the Financial Policy score at the score of the opco, or where there are multiple opcos, we cap it at a composite score, typically the average Financial Policy score of the opcos, weighted by the value of each ownership stake.

Notching factors

We consider the following notching factors for a minority holdco: (1) the structural subordination of lending at a minority holdco; and (2) the degree of influence or control the minority holdco has over financial and operational decisions at the opco level and the level of ownership.

Notching factor: Structural Subordination

After considering the rating or standalone credit profile of the opco and the scorecard outcome of the minority holdco, we consider the structural subordination of lending at a minority holdco.

Where the holding company holds only a minority interest in the opco, there are additional risks because a minority holdco generally does not fully control the opco or have access to the opco's liquidity reserves. Additionally, credit availability is typically more restricted and more cyclical for a leveraged minority holdco than for an opco or a holdco with full ownership and control.

For broad guidance on notching considerations resulting from structural subordination at a holdco, see our cross-sector methodology that discusses notching corporate instrument ratings based on differences in security and priority of claim.[13]

Notching factor: Degree of Influence or Control, and Ownership Level

For a minority holdco that we rate using this methodology, we typically make additional downward incremental notching after considering structural subordination based on (1) the degree of influence or control the minority holdco has over financial and operational decisions at the opco level and (2) the level of ownership.

To be rated using this methodology, a minority holdco typically has some representation on the opco board of directors and effective veto power over changes to the opco's distribution policy that would lower the distributions the minority holdco receives. We consider this to be moderate control. We typically consider a minority holdco to have substantial control when it has meaningful board representation or ownership rights that effectively provide it with co-control over important operational and financial decisions at the opco, including veto power over distributions, the ability of the opco to take on more debt and other material financial policies and corporate actions.

When a minority holdco has substantial control and owns a minority interest that exceeds 30% of the opco, we typically apply one additional downward notch from the standard downward notching for structural subordination. We typically apply a second downward notch if the minority holdco has only moderate control with more than 30% ownership, or substantial control with less than 30% ownership. When the holdco has moderate control and less than 30% ownership, we typically apply three notches of additional downward notching. Where there are minority stakes in multiple opcos, we typically use an average ownership level, weighted by the value of each ownership stake.

Exhibit 7

Additional downward notching based on degree of influence or control, and ownership level

		Ownership level	
		30 to 50%	15 to 30%
Degree of influence or control	Substantial	1 notch	2 notches
	Moderate	2 notches	3 notches

Source: Moody's Ratings

As an illustrative example, holdco A has a single minority stake: 35% ownership and moderate control in opco B, which has a rating of Baa3. After taking into account the minority holdco scorecard adjustments and other rating considerations described above, we may consider that the adjusted credit profile of opco B is Ba1. Using the guidance from the cross-sector methodology for notching corporate instrument ratings based on differences in security and priority of claim, we may apply one downward notch for structural subordination at the holdco level. We would then typically apply an additional two downward notches based on minority ownership (more than 30%) and the moderate control over decisions made at the opco level, which could lead to an indicative credit profile of B1 for holdco A.

Other considerations

For a minority holdco, in addition to the other rating considerations that apply to the opco, we consider additional risks and mitigants.

To assess liquidity at the minority holdco, we also consider the amount and timing of cash receipts and outflows, minority holdco liquidity reserves, and the potential for interruptions in the receipt of distributions from the opco.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

13 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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